Companion papers: **The Shape of Gravity** (the law) · **The Warp Drive** (the mechanism) · **The Hover Disk** (the materials) · **The Hexalock** (the vehicle) · **Antigravity Devices** (the bench floor) · **Superconductivity** (the coherence science)

Electrogravitics

An engineering field guide — from Townsend Brown's capacitors to a coherent-substrate drone: what is measured, what is derived, what is open, and how a drone company would actually get started.

draft 0.3 · 2026-08-18 · Osika Innovation AB · Alexander Osika, Claude

> **Claim boundary.** This paper makes no new physics claim. Its job is to be the single honest on-ramp to a scattered literature: it states, with the original numbers, (1) what the electrogravitics lineage actually measured — which in air is ion wind, fully explained by conventional physics; (2) what the published vacuum record bounds — every published, independently instrumented vacuum test is a null with quantitative bounds, while the one claimed vacuum positive (the 1955–59 French program, now read in the original: null on the Brown geometry, one uncontrolled ≈1 g·cm datum on a $BaTiO_3$ article) remains open; (3) what the companion papers derive — a specific force law with exactly one open number; and (4) the cheapest experimental program that would move that number, in the order that keeps a builder from fooling themselves. The coherent-coupling channel this program tests is an *unproven hypothesis, currently null on every substrate measured*. Nothing here licenses a product claim.

Every claim below carries one of six tags: MEASURED (instrument, published, or receipted in this project) · DERIVED (follows from the framework's law, receipt named) · ESTIMATE (a prior, honestly labelled) · PRE-REG (frozen prediction, not yet run) · OPEN (the bench decides) · NULL / REFUTED (tested and failed).

PART I — THE INTUITION

1. The word, and what was actually measured

Electrogravitics (that is the standard spelling) is Thomas Townsend Brown's word, from the 1920s, for the claim that a strongly asymmetric, high-voltage capacitor feels a net force toward its small electrode — and that this force is gravitational in character, not electrical. The lineage is real and documented: Brown's patents (GB 300,311 of 1928 through US 3,022,430 of 1962), the Project Winterhaven proposal to the military (1952), and the *Electrogravitics Systems* survey (Aviation Studies International, February 1956) that briefly made "electrogravitics" an aerospace buzzword. MEASURED

The capacitor is the part of Brown that reached the refereed literature, and it is all this guide needs. It is not all Brown did: his dated notebooks (1955–77) also carry a thirty-year "sidereal radiation" electrometer program, a rock-electricity program, and the provenance of every "time travel" sentence attached to his name — scored in the companion paper Townsend Brown.

Here is what six decades of instrumentation established about the force Brown measured:

- **In air, the force is ion wind** — electrohydrodynamic (EHD) thrust. Charge is sprayed off the sharp electrode, drifts to the blunt one, and drags neutral air with it. The thrust obeys `T ≈ I·d/μ` (current × gap / ion mobility, $\mu \approx 2\times10^{-4}$ $m^2V^{-1}s^{-1}$) — it scales with *current and gap*, not voltage, which is the fingerprint. Measured lifter thrusts reproduce this to within geometry factors of 0.1–0.8 (Bahder & Fazi, ARL-TR-3005, 2003 — often miscited as supporting electrogravitics; it is an ion-wind paper). MEASURED
- **Ion wind is genuinely useful** — MIT flew a solid-state airplane on it (Xu et al., *Nature* 563, 2018: 2.5 kg, 5 m span, 40 kV, ~6.25 N/kW). It is also intrinsically atmospheric: *zero thrust in vacuum*, and human-scale EHD lift needs ~330 m^2 of electrode per person. A real technology, the wrong tree. MEASURED
- **In vacuum, every published, independently instrumented test is a null.** Talley (AF-funded, 1988–1991, 10^{-6} torr): no thrust except during breakdown current. Tajmar (2004): the residual force was isolated corona wind. Bartlett & Tajmar (*Acta Astronautica* 181, 2021): ±0.3 mg resolution to 10 kV, no force in any configuration. The one claimed exception — the 1955–59 SNCASO "Project Montgolfier" vacuum residual — has now been read in the original: on a torsion balance at 5×10^{-5} mm Hg the bare Brown disc–wire article gave *nothing*, and the single positive (≈1 g·cm on a $BaTiO_3$-loaded capacitor at ±55 kV) has no current or breakdown log; see its §3 ledger row. NULL (published record)

The verdict, stated first because everything else depends on stating it first: Biefeld–Brown electrogravitics, as a *static* antigravity claim, is *refuted*. The force Brown measured in

> air is ion wind. (The transient/pulsed regime — where every positive in the record actually sits — is a different question, taken up in §1.1.) This paper's own framework does not fight that verdict — it *requires* it (§4): an incoherent capacitor is predicted to couple to spacetime at the bare Einstein-constant floor, $\sim 10^{-43}$, unmeasurably below every balance ever built. Any program that starts by disputing the nulls is not doing engineering. This verdict is scoped to what was measured: the atmospheric force — and the Montgolfier project's own 1959 report agrees, calling it "a pure demonstration of 'electric wind.'" The French *vacuum* residual is a separate, open question (§3 ledger): now read in the original — null on the Brown geometry, one uncontrolled ≈1 g·cm positive on a $BaTiO_3$ article. Only a replication with a current-and-force log — or the §6 bench — can move the tag.

So why is the word worth keeping? Because Brown got one thing right that the debunking, correctly aimed at his interpretation, never touched: **the geometry**. An asymmetric capacitor is a device for storing field energy with a built-in spatial gradient — a shape that stores energy on one side of itself. In the framework of the companion papers, that is precisely the shape a gravitational coupling would need (§2). Brown had the right blueprint and the wrong active ingredient: high voltage on an *incoherent* dielectric, where the framework says the missing ingredient is *coherence*. That diagnosis is falsifiable and cheap to test, which is what Part III is for. OPEN

Fairness to the man requires one more sentence. Brown was not a holdout against the ion-wind reading: in his own 1973 notebook re-appraisal he wrote that the torque of his instruments is "a direct function of the total current. There is no other energy source," that "perfectly dry transformer oil produces no torque," and that the 1920s belief that "the kinetic energy was somehow derived from the gravitational field" had been wrong; his 1958 theory even predicts that in vacuum, "with a K of unity … the force should be minimum." The static-vacuum-thrust claim is not in his notebooks; it is in his letters (1955, 1973 — from memory, "notes … not available to me now"), and the French torsion balance he cited was null on his own geometry (Townsend Brown §7). What he kept believing was that the *variation* of the force in time carried a signal — a different claim, scored separately there.

1.1 The verdict is about statics. The record is about transients.

Read the primary record with the static claim already set aside and one pattern is unmistakable: **every positive observation, from 1956 to today, is a transient one.** Brown's own vacuum note (Notebook §42, Apr 1956) is titled "The Impulse Effect": the force appears "concurrent with the vacuum spark," with the recovery of potential after it — not with the static field. Montgolfier's 1957 bell jar: motion "often precipitated by flashes," in two of four configurations sustained *only* by flashes; its 1959 balance moved only the barium-titanate article. Bahnson, Dec 1957: "noticeable attempt for the rig to rise momentarily" when the pulsator arced. Talley (1988–91): no force under DC, forces during breakdown and during pulsed drive through PZT — and a written recommendation to test higher pulsed voltages on high-K dielectrics that nobody has followed. Buhler (2023–26): dielectric stacks. Brown himself spent 1958 designing pulsed and RF drives (Notebook §§69, 72; the Bahnson "pulsator").

That is not a coincidence, and this framework says why: the retrodiction sim's evenness lemma (§6) makes an aligned *static* device sit at *exactly* zero, and it locates every possible net force in the rectified time-asymmetry of the drive — "swirl / spin / jerk." So the transient regime is not a loophole in the nulls; it is the one cell the framework's own force law points at, and it is the cell the published null program (all DC, all static) has not entered. Stated as the verdict should have been stated: **Biefeld–Brown is refuted as a static effect and open as a transient one.** OPEN

Now the discipline that keeps this from becoming a rescue. "Transient" is also exactly where the conventional candidate lives: a vacuum flash is a cathodic arc, and cathodic arcs eject metal plasma at $\sim 10^4$ m/s — the vacuum-arc thruster, $\sim$10–20 μN per ampere, i.e. $\sim 10^{-5}$ N·s of impulse per coulomb transferred, always pushing the assembly toward the anode side — which is the "always toward positive" of every report in the lineage (Martins & Pinheiro 2011; Cornillon's own 1957 "anodic effect" hypothesis is the same idea). And Brown's §42 contains one line that cuts against a pure dV/dt reading: the impulse appeared with the spark "and not when wholly due to a chopper in the external circuit." So the honest state is not "it needs rapid transients, therefore it works"; it is that the transient regime has never been run with the four measurements that separate a coupling from a discharge:

1. **Impulse per coulomb.** Log charge transferred per flash and impulse per flash on the same trace. Ejecta recoil gives $\sim 10^{-5}$ N·s/C and cannot exceed the arc-thruster figure by more than the usual factor of a few; a coupling has no reason to sit at that number. Brown's "thousands of dynes" per impulse, if it was ever real, would need coulombs of charge to be ejecta — testable in one afternoon.

2. **Mass.** Weigh the electrodes before and after; look for the deposited film. Ejecta leaves both.
3. **External chopper vs internal spark** at matched dV/dt — Brown's own control, which by his account came out for the spark. Repeat it with a current log; if a spark-free ramp of the same slope gives the same impulse, it is not discharge physics.
4. **Force per coulomb vs dielectric K.** Ejecta recoil depends on arc current, not on the dielectric; the record's one consistent lever (Montgolfier's $BaTiO_3$, Talley's PZT, Buhler's stacks) is that high-K matters. If impulse *per coulomb* scales with K, that is the first datum in the lineage that ejecta cannot cover.

Those four are now part of the §6 program. Until they are run, the transient regime is an open thread with a named conventional candidate — not evidence for the coupling, and not evidence against it.

1.2 From first principles: why the transient regime is the only one — imported from the energy lane

The project's energy-lane papers (The Zero-Point Machine, The VTA, The Optical-Cavity Device) reach the same conclusion from a different door, and their bookkeeping is worth carrying over verbatim, because it turns "it needs transients" from an observation into a derivation. Five statements, each with its receipt:

1. **Port locality.** Closed, symmetric configurations feel and output nothing; net effect exists exactly where a subsystem is *open* — where its circuit closes through something beyond itself (`PORT_LOCALITY_SYNTHESIS_2026-07-20.md`; Ampère/Lorentz sim: closed loop 0.033% agreement and zero self-force, open bridge 29% divergence at the ports). A charged capacitor rigidly carrying its own static field is a closed loop. That is the refuted half, said in the energy lane's words. DERIVED
2. **Adiabatic is quiet; abrupt is productive and paid for.** The charging-half identity: writing a quadratic ledger to standing displacement suddenly costs x^2 gross, exactly half dissipated; an adiabatic ramp waives the dissipated half (`oph_charging_half_identity_run.json`). The mainstream twin: the dynamical Casimir effect — change a boundary in time and the vacuum yields real quanta, with production set by *how abruptly* the boundary changes (Wilson et al., *Nature*

479:376, 2011; Rukan, Gulla & Skaar, PRL 2026 — the count is infinite only for infinitely fast removal). Slow boundary change: nothing. Abrupt: something, and the pulse pays for it. A vacuum flash is the most abrupt boundary change a capacitor can undergo. MEASURED (DCE) / DERIVED (identity)

3. **Rectification needs an asymmetric waveform.** The retrodiction sim's own law: $F(+\Delta) = -F(-\Delta)$ exactly, so an unrectified cycle averages to zero; net force needs a preferred direction in *time* as well as in space (§6; the "swirl / spin / jerk" clause). Symmetric AC on an asymmetric capacitor is therefore predicted null — and the record agrees: Bahnson's notebook, May 1958, "AC: none" on the tent rig, and small only when biased. A *spark-and-recharge cycle is a natural sawtooth*: nanosecond collapse, millisecond recovery. Brown's §42 puts the impulse "with the recovery of potential." That is a rectified time-asymmetry occurring by accident. DERIVED
4. **High-K is the coupling channel, not a curiosity.** The energy lane's one clean measured law is output ∝ capacitance (the 1/d slope race, only linear capacitive near-field coupling lands on −1.00; Moddel, Maclay). Coupling scales with C, and C scales with K. Montgolfier's 1959 conclusion — a high-permittivity dielectric "makes the phenomenon more vigorous," the $BaTiO_3$ article being the only one to move — and Talley's PZT and Buhler's dielectric stacks are that signature. And $BaTiO_3$ and PZT are sintered ceramics: the VTA paper's grain-boundary junction fabric, a disordered array of nm gaps, is what a strapped-on titanate block is made of. Offered as structural rhyme, not receipt. OPEN
5. **The small input is the grid, not the fuel.** In the VTA reading nothing is amplified: the drive breaks symmetry and clocks the device; the output *format* follows the drive. Transferred here: the pulse is not the thrust's energy source in any exotic sense — the parametric route is a *transducer*, conservation applies in the ordinary way, and the figure of merit is impulse per joule of pulse energy at a given edge rate. "Propellantless, not reactionless" (§2): the reaction partner is the medium the open port closes through, or it is ejecta — which is what the discriminators decide. DERIVED

What this does to Brown's own control. §1.1 kept the honest caveat that Brown saw the impulse with the spark "and not when wholly due to a chopper in the external circuit." Under (2) that is not a refutation of the transient reading; it is a *rate* datum. A 1956 mechanical or tube chopper gives an edge of microseconds to milliseconds; a vacuum spark collapses in nanoseconds. "Spark yes, chopper no" is exactly what production-scales-with-abruptness predicts if the effect lives above the chopper's edge rate — and it is what ejecta recoil predicts too, since only the spark ejects metal. So the two readings separate on one axis Brown could not scan and we can: **discriminator 5 — impulse versus edge rate at fixed transferred charge, spark-free**. Modern solid-state HV pulsers (nanosecond edges, no arc) put a controlled dV/dt on the same asymmetric article without a discharge; ejecta gives nothing without the arc; a boundary-abruptness coupling gives an impulse that climbs with edge rate. Added to §6.

> **The sentence the field guide should have carried from the start:** the static Biefeld–Brown effect is zero by port locality and by measurement; the transient Biefeld–Brown effect is the one configuration in which the project's own physics — the evenness lemma, the charging-half identity, the capacitance-coupling law, and the dynamical-Casimir precedent — permits a net effect at all; every positive in the seventy-year record sits in that configuration; and it has never been tested against its conventional twin with the five measurements that would tell them apart. That is a target, and Brown's geometry — an asymmetric high-K capacitor under an abrupt, rectified drive — is the right first article for it. Register: OPEN, and it stays there until the bench speaks.

2. Sixty seconds of gravity, and what "engineering it" would mean

Start with the part that is not a hypothesis: **electromagnetic fields curve spacetime.** That is Einstein–Maxwell — coupling $8\pi G/c^4$, no free parameters — and it has been solved *exactly* for laboratory hardware: Füzfa (*Phys. Rev. D* 93, 024014, 2016) computes the full nonlinear spacetime curvature of real current-carrying coils (effect $\propto I^2$, i.e. $\propto B^2$), with a proposed interferometric detection. A field-conditioned machine *does* modify local geometry, unconditionally. THEOREM The honest arithmetic that goes with the theorem: Füzfa's CERN-magnet-class stack shifts an optical phase by 1.6×10^{-25} rad per round trip (10^{-11} rad only after 200 days of coherent integration), and at 100 T the Melvin curvature radius is 3.7 light-years. The mechanism is settled; the entire engineering question is magnitude — the ~40-order climb from the bare coupling to a useful one.

The mechanism story is owned by The Shape of Gravity; here is the engineer's summary, with the receipts named. In this framework spacetime is a network of elastic edges, and a mass is stable energy settled into it. One derived law with no free parameter — `F = 1 + E/E★`: the effective length of an edge grows linearly with the energy it carries — turns out to close all of weak-field gravity. Settled energy lengthens nearby edges (space curves) and slows clocks on them (time dilates), the same δ doing both, which forces Einstein's full factor-two light deflection with no tensor added: the ray-trace converges to 2.000 (`crypto_oracle/shape_F_gravity_closure_probe.py`), and the 1/r Newtonian halo fits at $R^2 = 0.9998$. DERIVED What is *not* derived is the absolute strength — Newton's G — and the paper says so on purpose. OPEN

> **The claim has three layers, and only one of them is a bet.** (1) Mechanism — theorem-level, unconditional: fields gravitate. (2) Bare-coupling magnitude — settled, and bracketed from above by the null program (§3). (3) Coherent enhancement — the open bet the §6 bench decides. The correct register is therefore "**this works; the climb is the program**" — exactly as fusion's question is confinement, not whether fusion works. The closest respectable published cousin of layer 3 is Ummarino & Gallerati's superconductor gravito-Maxwell work (EPJ C 2017 → Front. Phys. 2022: transient $\Delta g/g \sim 10^{-7}$–10^{-5} near T_c — screening, not generation, untested), which shows a peer-reviewed literature reaching for the same lever at far more modest magnitude.

If gravity is "settled energy sets effective geometry," then engineering gravity means one thing: **store a lot of coherent energy in a body, asymmetrically, so that the body sits in a geometry gradient of its own making — and couple to the lattice well enough that the gradient does mechanical work.** Three consequences follow immediately, and they organize everything a builder needs to know:

1. **Hover is statics; travel is dynamics.** A static force that balances weight does no work ($P = F \cdot v = 0$). If the coupling exists at all, holding position costs only losses — like a table holding a book, or a maglev rail holding a train. Accelerating costs real energy, and a conservative coupling makes deceleration regenerative. The lawful summary is "hover ≈ free, momentum costs work" — *not* "free energy." DERIVED
2. **Propellantless is not reactionless.** Every force needs a reaction partner. A propeller throws air; a rocket throws mass; this coupling, if real, hands momentum to the lattice — the same bookkeeping as maglev, where the rail takes

the reaction. "Where does the reaction momentum go?" is the first question to ask of any device claim, ours included. A body that rigidly carries its own static field configuration is a closed loop and gets exactly zero net thrust — confirmed to 10^{-17} N in simulation under both classical force laws (`crypto_oracle/universe_sim/ampere_tension_sim.py`). You cannot fall into a pocket you carry. DERIVED

3. **Symmetry gives zero.** A symmetric energy store makes a symmetric dent and no net force. All thrust lives in the asymmetry — in the drive's spatial gradient (the Brown geometry), or its chirality (swirl/spin), or its time-asymmetry (transient "jerk" drive). This is why the sign-reversal test (§6) is the cleanest discriminator: flip the asymmetry, and a real coupling flips sign; every artifact doesn't. DERIVED

> **"Lock, not lift" — the phrase a drone company should internalize.** A propeller *lifts*: it pumps momentum downward continuously, and when it stops you fall. A working coherent-coupling disk would *lock*: engage the coupling and the lattice pins the pose, at near-zero power, the way a rail pins a maglev. Thrust is then a controlled *modulation* of the lock — continuous through zero, sign set electronically in microseconds, no reversal transient, regeneratively recoverable. Every one of those properties is exactly what 6-DOF multirotor control wants and cannot have with propellers (§5).

How to picture it — five images that hold, and three that don't

A century of this literature has produced a handful of genuinely good mental pictures and a handful of seductive wrong ones. Both are worth having, because the wrong ones are the ones your team will meet first.

The images that hold. *Gravity does not pull; it adjusts the characteristics of space* — the best one-line statement of general relativity in the whole corpus, and it comes from the proponents' own 1956 *Electrogravitics Systems* report. *There is no perfectly rigid rod*: near a mass, rulers shrink and clocks slow, so a circle's circumference comes out less than 2πr — that is what "curved space" operationally means (Dicke's and Puthoff's medium picture, and it is mainstream). *Light bends near a mass the way it bends entering glass* — spacetime as an optical medium of varying index, a picture as old as Eddington. *A moving mass makes a gravity analogue of a magnetic field* — the spinning Earth drags spacetime like a spoon through honey, measured by Gravity Probe B at ~39 milliarcseconds per year, which honestly sets the scale of what mass currents do. And the one that governs the ledger: *the ion lifter is an invisible propeller made of air* — the wire is the blade you cannot see, each ion is a runner in a bucket brigade handing momentum to neutral air, which is exactly why the thrust scales with current, ignores which way the voltage is wired (a fan doesn't care which way the motor is wired), and dies dead in vacuum: no one to hand off to.

The images that mislead — teach them, then break them. *matter is at bottom an "electrical condition," matter is connected with gravitation, therefore electricity is too* (Brown's 1929 syllogism, paraphrased; the ancestor of every argument since): the premise is true — mass-energy includes field energy, which is precisely why a charged capacitor weighs imperceptibly more — but "connected at 10^{-43}" is not "controllable," and the sentence smuggles one into the other. *The surfboard on a self-made hill* (Mason Rose, 1952; the single most memorable image in the field): a real surfer rides a slope sourced by the Earth and maintained by the ocean; a craft that makes its own hill and carries it along is pushing its car from the inside — the reaction has to land somewhere, and if that somewhere is the air you have built an ion thruster. Our own conservation books say the same thing in equations: a symmetric, carried pattern gives exactly zero (§4). "*Ion wind is three orders of magnitude too small*" — a half-quotation of the Army Research Lab paper whose *other* half computes ion drift at the right magnitude, after which NASA (2004) and Ianconescu (2011) turned it into a full model that predicts the thrust, its ground-wire dependence, and its polarity independence. And the one image to keep from the proponents because it is almost right for the wrong reason: the 1956 report's *electrostatic wing* — "the same lift as pressure and suction over a wing, except that no airflow is necessary." Right shape; the one wrong clause is the entire lesson.

3. The measured ledger

One table, so nobody has to reconstruct the evidentiary state from folklore. "Bound" is the constraint the measurement places on the framework's one open coupling number (§4).

Item	Record	Status
Ion wind (EHD) thrust in air	$T \approx I \cdot d/\mu$; lifters, MIT 2018 airplane; ~0.15 mN/mA at 3 cm gap	MEASURED — real, atmospheric-only, conventional

Biefeld–Brown force as antigravity — *static / DC*	Explained by EHD in air; vacuum nulls below; and by this framework's own evenness lemma an aligned static device sits at exactly zero	REFUTED (static)
Biefeld–Brown force — *transient / pulsed / high-K* regime (§1.1)	Every positive in the primary record is transient: Brown's 1956 impulse at the vacuum flash; Montgolfier's flash-driven rotations and its one 1959 positive on a $BaTiO_3$ article; Bahnson's arc-moment rises; Talley's breakdown/pulsed-PZT forces (with his unfollowed recommendation to test higher pulsed voltage on high-K); Buhler's dielectric stacks. Conventional candidate: vacuum-arc ejecta recoil ($\sim 10^{-5}$ N·s per coulomb, always toward positive). No published test has run the discriminators (§1.1)	OPEN THREAD — the untested cell, and the one this framework's own force law points at
ONR Pasadena File 24-185 (Cady, 15 Sep 1952) — the only calibrated measurement of a Brown saucer, made on Brown's own apparatus in his presence	Two 12-in Al / 18-in Plexiglas discs on a 6-ft rotor: ~7 g thrust at ~30 kV, ~1 g with the corona wire shrouded, ~3 g on polarity reversal (ratio 2.5 vs ion-mobility ratio 1.9); electric-wind estimate "about twice the observed" — "satisfactory in order of magnitude"; efficiency ~1.4%; "presumably … zero in a vacuum." Also: "The apparatus appears to be well engineered and extremely reliable."	MEASURED — ion wind, at the source
Talley 1988–91 (USAF), 10^{-6} torr	No thrust except during breakdown current	NULL
Tajmar 2004, 38 kV, corona-excluded	$<10\ \mu N \rightarrow$ coupling $< \sim 2\times10^{-3}$	NULL
Bartlett & Tajmar 2021, ≤10 kV	nano-N resolution, no force → coupling $< \sim 2\times10^{-10}$ — **the tightest published bound**	NULL
1955–59 Paris tests ("Project Montgolfier", SNCASO/Sud Aviation, facilitated by J. Cornillon;	**Read in the original (2026-08-18; receipt `MONTGOLFIER_REPORT_RETRIEVAL_2026-08-18.md`).** Its own 1957 report attributes the *air* motion to a non-polarised electrode ("tourniquet"/electric-wind) effect — agreeing with this ledger — and its control turnstile that spins at 40 kV in air does not turn at all to 150 kV in vacuum. The 1957 bell-jar rotations were polarity-	OPEN THREAD — one numbered positive on one $BaTiO_3$ article, uncontrolled for current; null on the

reports 1 Jul 1957 and 15 Apr 1959)	locked (always toward the positive pole), often flash-driven, killed by a grounded screen, drew 2–4 μA at 40 kV, and could not be measured. The **1959 final report** (1.48 m steel chamber, 5×10^{-5} mm Hg, torsion balance, ±50–55 kV): the disc–wire (Brown) article **nothing**; two discs in plexiglas **nothing**; araldite-sphere capacitor **nothing**; plexiglas plane capacitor 3–4° oscillations; **only** a $BaTiO_3$-loaded encapsulated capacitor gave 30° ≈ **1 g·cm torque**, reversing with polarity — no lever arm, current log, or breakdown log recorded (a 10–20 cm arm would mean ~0.5–1 mN). Conclusion in the report: high-K dielectric "makes the phenomenon more vigorous." Conventional candidate for the flash-driven part is named in the report itself (Cornillon's "anodic effect" = charge-emission momentum; cf. Martins & Pinheiro 2011)	bare Brown geometry in the same run
Buhler / Exodus Propulsion (asymmetric dielectric capacitor stacks)	Patent's one measured number (237 μN) was in air — inseparable from ion wind; the mN-scale vacuum figures are talk-claimed (~2000 runs), unreviewed, unreplicated — and declining: the claimed force fell ~40× between Apr 2024 (1 g on a 30–40 g device) and Mar 2026 ("5–10 mN," rebranded "electrostatic pressure force") with no published explanation — the monotonic decline-under-scrutiny signature — still with no paper and no independent instrumented test, while the claim sits 10^5–10^6× above the Dresden noise floor in the same HV regime. Back-solving our force law on the claimed numbers implies a coupling ≈ 2×10^{-4} — consistent within an order of magnitude, *not validated*, and ~125× below the hover threshold	OPEN THREAD — watch, don't bank; declining
Tajmar, Kößling & Neunzig 2024 (*Sci. Rep.* 14:19427): broad gravity–EM coupling sweep	Capacitor geometries, solenoids, tunneling currents, varistors, ε/μ gradients, crossed & helical B, shielded toroids — all bounded at **~nN force / nN·m torque** in high vacuum; supercurrent→gravity bounded at 5×10^{-9} N/A (*Front. Phys.* 2022). The tightest map yet of which shortcuts do not exist	NULL (comprehensive)
Podkletnov rotating-superconductor weight loss	NASA Marshall and Göde replications null to 0.001%	UNREPLICATED

Li–Torr gravitomagnetic amplification (theory)	Refuted on its own terms (Harris 1999); AC Gravity LLC published nothing	REFUTED
Graham et al. 2008 (Canterbury), rotating s-wave SC ring laser	The cleanest existing upper bound on lattice coupling in ordinary superconductors	MEASURED bound
Pais "HEEMFG" (Navy, 2016–19, ~$500k)	"Could not be proven" — but the rig could spin *or* vibrate, never both, and never reached the specified charge: an uninterpretable null, not a refutation. Mine the patents for geometry only (§7)	NULL (weak)
Wallace patents US 3,626,605/606 (GE era, 1971): rotating half-integer-nuclear-spin masses	13-arcmin torsion claim; unreplicated — and, unlike Brown, **no published null either**	OPEN THREAD (§8)
Spinning-mass weight change (Hayasaka & Takeuchi 1989, ~10 mg on a right-spinning gyroscope)	Refuted within a year by three independent groups — Faller et al., Nitschke & Wilmarth, Quinn & Picard (all 1990), the last ~30× below the claimed effect. Rotating-mass weight anomalies are bounded at ~10^{-7} of the mass; *counter*-rotation adds a symmetry, not an effect. This single row bounds every "spin it and it gets light" saucer design (Carr, Searl, the Repulsine tonnage claims) at once	NULL (published record)
Carr OTC-X1 / Utron (1957–61) and its 2021 revival (Gobbi)	Patent is an amusement ride; prototype failed on film under external drive (mercury leak); "580 rpm" decodes as rim speed ≈ Earth's equatorial 465 m/s (0.35% of g of relief; Al hull 1.7× past yield); Utron ≈ 8 V mercury homopolar disc; the 10^{13} W / 10^{5} T revival rests on a 10^{14}× current error. Full audit: Otis T. Carr and the OTC-X1	REFUTED / PRICED
This project's own bench	No substrate measured to date has shown the coupling; every xv-channel measurement so far is null	NULL to date

Read the last row again before reading Part II. The framework's force law is a *prediction under test*, made by people who publish their own nulls. That is its credential.

PART II — THE ENGINEERING FRAME

4. The force law, for engineers

The mechanism paper (The Warp Drive) and its math audit (`WARP_DRIVE_MATH_AUDIT.md`) give the working force law. As pressure on a driven substrate of area A:

```
P_w  =  C·κ·½ ε_r ε₀ E²        [Pa]         (field / capacitor drive)
F    =  C·κ·U_drive / L_grad    [N]          (general stored-energy form)

C  ≡  C_sub·|χν|²  — the lattice coupling            ← THE open number
κ  ∈ [−1,1]        — drive asymmetry (0 = symmetric = zero force; sign(κ) = sign(thrus
U_drive            — coherently stored energy (not dissipated power)
L_grad             — gradient length ≈ device radius
```

> **What C actually is — the shout and the room.** Think of the drive's stored energy as a *shout*, and the lattice — spacetime's substrate — as a *room that can echo*. How much force comes back depends on two independent things, and that is why C is a product of two factors. **$|\chi\nu|^2$ is the tuning**: how well the substrate's own coherent mode overlaps the lattice's mode — matched frequency, matched geometry, phase locked across the whole body. It is dimensionless, sits between zero and a hard geometric ceiling (the derived vertex-sharing bound, $g \leq 1/\sqrt{3} \approx 0.577$), and enters squared because coupling is an amplitude and force follows energy — the same reason a wave's power goes as amplitude squared. A bronze cymbal, driven at audio, tunes at $|\chi\nu|^2 \sim 10^{-14}$; an optically driven diamond disk is estimated near 10^{-1}. **C_sub is the room's echo efficiency**: given that the modes overlap at all, what fraction of that overlapping energy the *substrate* converts into force — how much of the body participates coherently, how much energy it can hold before it cracks or breaks down, how consonant its structure is with the lattice. It is the honest name for everything about the *material* that is not the mode overlap. They are kept separate because they fail independently and are fixed independently: perfect tuning on a body that cannot hold energy is nothing, and a superb energy store driven at the wrong frequency is nothing — force needs both, so the materials program is two ladders that multiply. One number is what a balance reads out; splitting it into two is the framework's claim about what that number is made of, not a

> second measurement. And the scale to hold in mind: an incoherent lump — Brown's capacitor — sits at the bare-vacuum floor of C, the Einstein constant $8\pi G/c^4 \approx 2\times10^{-43}$, while hover needs $C \approx 10^{-6}$. That forty-order climb is the whole job, and both factors are levers.

Hover is the pressure condition `P_w ≥ ρ·t·g` (areal weight) — and because area cancels, **self-lift is size-independent**: a property of material and drive, not diameter. Scale anchors, all DERIVED from the law plus known material constants:

- A 1 mm diamond-class skin weighs ∼**34 Pa**. Full electrostatic pressure at 1 GV/m in diamond is ∼25 MPa — so hover needs the coupling·asymmetry product to deliver only $\sim10^{-6}$ of the stored pressure. On the coherence dial of the audit: **hover at S_coh·f_cons ≈ 1.4×10^{-6} — about one part per million of full coherence.**
- The consonance ceiling is f_cons ≤ e^(−P/12) ≈ 0.873; the bare-vacuum floor — what an *incoherent* body couples at — is the Einstein constant $8\pi G/c^4 \approx 2\times10^{-43}$. The whole game is how far real coherence climbs a body up those ∼40 orders of magnitude. OPEN
- The tightest published null (Bartlett & Tajmar 2021) bounds *statically driven* capacitors at $C < 2\times10^{-10}$ — which is 33 orders above the floor and 4 orders below Buhler's implied value. The framework's own explanation for why the nulls don't close the question: every null device to date was a static, incoherent capacitor, which by the framework's own predicates cannot access the coherent channel. That is self-consistent, and it is also the framework rescuing itself with its own predicate — the papers say so plainly, and the §6 program is designed to close exactly that loophole rather than lean on it. OPEN

4.1 The levers

Lever	What it is	Status

Stored energy U_drive	Coherently stored, not dissipated. Material ceiling is specific resilience $R = \sigma^2/(E\rho)$ for mechanical storage, $\frac{1}{2}\varepsilon_r\varepsilon_0 E^2$_bd for field storage. Diamond is the single-material champion on both counts; graphene's ceiling is ~7000× higher again but its coupling velocity is genuinely uncertain	DERIVED
Asymmetry κ	The Brown geometry: small/large electrode, domed top over larger bottom shell, insulation ring — an asymmetric capacitor is κ rendered in hardware. The asymmetry "wants to be a swirl": chiral electrodes, spin, and time-asymmetric (jerk) drive are the same lever in space, rotation, and time	DERIVED (form) / OPEN (magnitude)
Coherence S_coh	The claimed missing ingredient. Fraction of the body participating in one phase-locked mode: Q, isotopic purity (^{12}C, ^{28}Si), condensate boundaries, cold. See Superconductivity for the measured science of macroscopic phase coherence	OPEN
Consonance f_cons	Geometric match between the substrate's mode pattern and the lattice (icosahedral symmetry, φ-ratios, degree-3 carbon cages, quasicrystals). Ceiling 0.873	OPEN
Frequency	**Deliberately listed last, because it is the fork, not a lever.** Hypothesis A: coupling rises with drive frequency ($\eta \approx v\_phase/c$ — THz/optical drive wins by 8–12 orders). Hypothesis B: coupling is stored-field-limited and DC suffices (Buhler's DC claims point here). A frequency sweep on one device decides it (§6). Do not design a program that assumes either answer	OPEN — the A/B fork

What a null bounds, and what it doesn't. A vacuum null on your device bounds $C < \delta F \cdot L\_grad/(\kappa \cdot U\_drive)$ for *your* drive condition. It refutes the mechanism only if the chamber-variation and frequency-sweep controls also come back flat. And know exactly what the published null program covers: it — including the comprehensive 2024 Dresden sweep — is excellent metrology, and it is binding *for what it tested*: static fields, DC currents, incoherent substrates. No published test article has carried even one of the four §4.1 coherence predicates (lattice-matched symmetry, coupling channel, condensate boundary, near-mode drive), and a static — or uniformly rotating, mirror-symmetric — drive has κ = 0 by the conservation books: it structurally cannot rectify. This is worth stating as a two-

> framework agreement: for every article the null program tested, general relativity predicts effectively zero (weak-field arithmetic) *and this framework predicts exactly zero* (symmetric drive, κ = 0, closed loop) — the same weak-field regime our companion law re-derives. The two theories part company only where the coherence predicates are met and the drive is rectified; nowhere tested so far is in that regime. So the test matrix has exactly one untested cell, and the coherent-substrate architecture of the companion papers sits in it. That is not a rescue; it is a target. The §6 bench is the first experiment that would fill the cell, and a null *there* — with the coherence predicates instrumented and verified present — would be the first null that actually reaches this framework. Conversely — and this is the discipline that keeps you honest in the other direction — a *positive* force on your balance is an artifact until it survives: vacuum, thermal null, dummy-load null, sign reversal under drive flip, chamber variation, and sidereal folding. The published field is a graveyard of people who skipped one of those.

5. Why a drone company is the right builder

It is tempting to think this technology, if real, belongs to aerospace primes. The opposite is true, and the reasons are concrete:

1. **The control problem is already solved, and it's yours.** A craft with six independently throttleable force patches is a 6-DOF wrench-map inversion — exactly the mixer a multirotor autopilot runs today. The Hexalock paper works this out on the published Lynchpin airframe (Howard et al., SAE Int. J. Aerosp. 16(3), 2023 — a 651 g prototype that *flew*, on PX4, with six bidirectional props): six disks on the six pentagons of a collapsed dodecahedron, a 6×6 always-invertible mixer, one RP2350 whose 12 PIO state machines map one-to-one onto 6 disks × {amplitude, κ-steer}. Swapping the thruster type does not touch the mixer. The flight-control ledger is closed; only the lift ledger is open — and a drone company is precisely the builder for whom the closed half is free. MEASURED (the airframe/control half)
2. **The disk fixes your hardest actuator problem.** 6-DOF vehicles need signed, arbitrarily small thrust per axis — the constraint that pushed ETH's Omnicopter to eight over-actuated rotors and the Lynchpin prototype to bidirectional props with reversal latency. A coupling disk's thrust is continuous through zero with electronically-set sign: no mechanical state, no reversal transient. DERIVED (from the law's κ-linearity — conditional on the coupling existing at all)

3. **Endurance inverts.** A propeller drone sizes its battery for hover; a lock-mode craft sizes it for *maneuvering*, because station-keeping costs only losses. The first real product isn't a faster drone — it's a drone that holds station indefinitely on a trickle. DERIVED (conditional, same caveat)
4. **The bench is drone-shop-sized.** Nothing in Part III needs a wind tunnel or a cleanroom: a vacuum bell jar, a torsion or pendulum balance, a HV supply, an RP2350, and discipline. The single most dangerous item is the 40 kV supply, which drone shops handling LiPo banks already respect.

> The strategic shape: everything above the coupling is flight-proven engineering you already do; everything below it is one number. Build the program that measures the number, and structure it so that every stage yields a sellable conventional capability (ion-wind actuation, HV competence, precision force metrology, THz materials) even if the number stays zero.

PART III — THE PROGRAM

6. Getting started: the bench program

Rule 0: bench one disk before you build six. The Lynchpin paper states it as a failure mode: the headline risk is that the disk does not lift at all, so never invert the order. Everything below is one disk on one balance.

Stage 0 — Calibrate on physics that works (weeks, ~$1k)

- **Build the ion lifter first.** Fly it, measure T vs I·d/μ, watch it die in the bell jar. Now your team owns the artifact that contaminates this entire field, and your instruments are proven on a real force. MEASURED
- **Diamagnetic control:** pyrolytic graphite levitating over NdFeB — zero-power levitation on textbook physics ($B \cdot dB/dz = \mu_0 \rho g/|\chi|$). This pins the conventional baseline any later in-field force claim must be distinguished from. MEASURED
- **Balance floor:** a 2 m pendulum or torsion balance with interferometric readout reaches ~nN–500 pN. Characterize thermal drift, EMI, and electrostatic pickup *before* any candidate device goes on it.

Stage 1 — The discriminator set (the actual experiment)

One asymmetric capacitor device, in vacuum, on the calibrated balance, run through five controls. This set — not any single reading — is the experiment:

1. **Vacuum arm, always.** No in-air number means anything in this field. (The audited-and-rejected proposals share one tell: no vacuum control anywhere in the plan.)
2. **Sign reversal under drive flip.** Flip the asymmetry (polarity / phase / chirality): a lattice coupling reverses sign; heating, outgassing, and electrostatic pickup do not. The cleanest single discriminator the framework offers.
3. **$\kappa \to 0$ null.** Symmetric drive at equal power must give zero. If it doesn't, you're measuring an artifact.
4. **Frequency sweep, DC → RF (→ THz where you can).** This is the A/B fork of §4.1 decided on your own hardware: force rises with frequency ⇒ hypothesis A; flat ⇒ hypothesis B (or nothing).
5. **Chamber variation + sidereal folding.** A lattice coupling is chamber-independent; trapped charge tracks the walls. And fold every long log at both 24 h 00 m and 23 h 56 m: any real preferred-frame coupling modulates at the sidereal day, where no mundane artifact lives. Costs zero hardware on a timestamping MCU.
6. **The transient-regime discriminators (§1.1).** Because every positive in the record is a flash, pulse or breakdown event, log *impulse per coulomb* on the same trace as force (ejecta recoil sits at $\sim 10^{-5}$ N·s/C); weigh the electrodes before and after; run Brown's own external-chopper-vs-internal-spark control at matched dV/dt with a current log; take force *per coulomb* against dielectric K; and (fifth, from §1.2) sweep *edge rate at fixed transferred charge* with a spark-free solid-state nanosecond HV pulser — ejecta needs the arc, an abruptness coupling needs only the edge. A pulsed positive that survives these five is the first datum in the lineage that discharge physics cannot cover; one that fails them is a vacuum-arc thruster, which is a fine thing to have built and not what anyone was looking for. The pulse energy is the budget: log impulse per joule as well as per coulomb, so the parametric reading is priced honestly as a transducer.

Any force that survives all six is worth a phone call. A null across the sweep is also a result: it is the tightest bound ever placed on the coherent channel, publishable as such.

OPEN — this is the bench that decides

Stage 2 — The first article worth building (months)

The corpus's convergent first build ("Architecture C") is deliberately shop-buildable: two facing domed copper discs, 80–120 mm, across a corrugated gap; a **chiral-spiral electrode** charged to a few kV (the spiral *is* the κ direction — no steering electronics); the shell **spun** by one balanced BLDC at governed RPM (a flywheel, not a propeller — spin supplies the high-frequency mode mechanically, which no published null device had); per-disk flyback or Cockcroft–Walton HV; passive cooling, because colder is strictly better for coherence. Rationale: every published null was a *static* capacitor; this device is the cheapest object that actually exercises the claimed channel (stored field + asymmetry + rotation + time-structure) with independent control of each.

OPEN — a build hypothesis, not a result

Stage 3 — The coherence ladder (the materials program)

Owned by The Hover Disk; the drone-relevant summary: substrate ladder steel → bismuth → $BiFeO_3$ → quasicrystal; the single-material optimum is a ^{12}C-enriched single-crystal diamond disk driven optically (structural champion *and* $\eta \approx 0.41$ optical substrate); the layered endgame adds an icosahedral-quasicrystal symmetry layer, a THz-coupling multiferroic film, and a superconducting boundary (YBCO at 77 K) as the phase-coherence layer. Two free pre-screens: broadband absorptance (good couplers should be anomalously black — necessary, not sufficient) and the diamagnetic control from Stage 0.

> **Superconductor discipline.** The condensate boundary enters as a *coherence layer*, not as an antigravity ingredient. The measured record is unambiguous: flux expulsion is a property of the superconducting state, not a road into it, and every rotating-SC weight-loss claim is unreplicated (Podkletnov) or refuted (Li–Torr). The honest statement is Canterbury 2008's: ordinary s-wave superconductors bound the coupling from above. If a condensate boundary matters here, it is because phase coherence is the claimed missing ingredient — and that is exactly what Stage 1 measures.

The desk work that grounds it

The Biefeld–Brown retrodiction sim — predictions frozen in a pre-registration before the run (`papers/BIEFELD_BROWN_SIM_PREREG_2026-07-19.md`), then executed (`crypto_oracle/universe_sim/biefeld_brown_sim.py`, receipt `runtime_receipts/biefeld_brown_run.json`, results `papers/RESULTS_BIEFELD_BROWN_2026-08-13.md`) — **passed all four frozen predictions and all three gates.** An incoherent Brown-class capacitor at the bare-vacuum floor produces at most 3×10^{-42} N in vacuum — 35–39 orders below the Talley/Tajmar/Bartlett balances, retrodicting every published null; the in-air force is reproduced by the ion-wind side-calc to within 12% with zero metric contribution; the coherent arm crosses the hover threshold at $C \approx 1.6\times10^{-6}$ (the audit's hand-derived dial value was 1.4×10^{-6}, frozen before the run); and the conservation controls (symmetric drive, carried pattern, static alignment) are zero to machine precision — the bootstrap is structurally impossible in the model, not merely penalized. MEASURED (in-model) Per the pre-reg's own register rules: this validates that the framework's story is *coherent and consistent with the published nulls* — it is not bench evidence that the coherent force exists. The bench (§6, stages 0–2) still decides reality.

7. Dead ends, priced — do not spend money here

- **Unruh / zero-point-field propulsion.** The one engineering proposal in the gravity paper's audit whose premise is true (the Unruh bath is textbook physics) was priced with every assumption in its favour: $T = 4.06\times10^{-21}$ K per m/s² — so no vibrating structure of any kind is a route (a 40 kHz × 10 μm ultrasonic drive makes 2.6×10^{-15} K of bath); break-even sits at 160.7× the Schwinger field, i.e. *the vacuum tears before the bath pays for itself*; and the technology-proof killer is the identity $\tau = 2\pi c/a$ — the bath's coherence time is set by the same acceleration that makes it, so there is no configuration in which you aren't paying for the reservoir you're harvesting. PRICED & CLOSED (`runtime_receipts/unruh_zpf_anisotropy_pricing.json`)
- **Free-air acoustic hover.** The 1-atm ceiling and the momentum bookkeeping exclude human-scale free-air acoustic levitation — a pressure cushion's reaction goes into the ground, so it is ground-effect only. (This project retracted its own

hoverbike lift story on exactly this audit; the corrected diagnosis is in `hover/GROUND_EFFECT_DIAGNOSIS.md`.) EXCLUDED

- **The over-unity package deal — stated precisely.** What conservation rules out flat is *closed-system* over-unity: cyclic net work from a single equilibrium bath, no reservoir named. Devices offering lift *plus* that are the single most reliable fraud marker in this literature — refuse the package. FORBIDDEN What this does *not* cover: open-system vacuum/near-field energy harvesting, which is a separate lane in this project with a measured, peer-reviewed anchor (Moddel-class MIM optical-cavity devices, ~70 W/m², output ∝ 1/d) and a lawful frame — rectifying the ambient thermal near-field against the effectively-0 K zero-point sector is a Carnot pair, and its non-negotiable signature is that a genuine device **runs cold under load** while an artifact warms. See `zero_point_energy.html` §7 and `optical_cavity_zpe.html`. The discipline: never argue conservation by reflex; *demand the reservoir inventory and the thermal sign*. A device claiming lift + energy that names its reservoir and shows the cold signature has earned a bench date, not a dismissal — one that names neither gets the fraud-marker treatment. OPEN (energy lane's question, not this paper's)
- **Pais patents as mechanism.** The Navy patents are real and granted; the claimed physics fails by ~10 orders on its own numbers, the HEEMFG program's null stands (weakly), and the RT-superconductivity patent fails a kinematic bound a quartz watch refutes. Mine the *geometry* (double-shell charged/insulated cavity, counter-spin, acousto-optic dual drive) and discard the claims. REFUTED as physics / OPEN as geometry
- "**The strong force is amplified gravity**" (Nuclear Gravitation Field Theory and kin): fails universality — the dineutron, which its own logic predicts binds tighter than the deuteron, is not bound at all. REFUTED
- **Otis Carr's OTC-X1 / Utron (1957–61) and its 2021 revival.** Audited in full in the companion Otis T. Carr and the OTC-X1: the patent is an amusement ride, the prototype failed on film under an external motor (mercury leak), the "580 rpm" decodes as rim speed = Earth's equatorial speed (0.35% of g of relief; aluminium hull 1.7× past yield), the Utron is a ~8 V mercury homopolar disc, and the 10^{13} W / 10^5 T revival rests on a 10^{14}× current error. Counter-rotation is this framework's own zero case. Residues: liquid-metal homopolar machines and self-excited dynamos are real — as loads, not sources. REFUTED / PRICED

8. The under-exploited threads

Three places where the next real datum is cheapest:

1. **The Wallace rerun.** Henry Wm. Wallace (GE, patents US 3,626,605/606, 1971) built rotating-mass devices whose materials were chosen by one axis nobody else has used: *half-integer nuclear spin* (Cu-63/65, Zn-67, Pb-207; spin-zero Fe and ^{12}C as "insulators"). Unreplicated — and with no published null either. His Barnett-effect nuclear polarization at 28 kRPM was $\sim 10^{-10}$; modern hyperpolarization (DNP/SEOP) reaches 0.1–1 — **a rerun today is nine to ten orders of magnitude stronger on the source side**, at university-lab cost. Bismuth-209 (spin 9/2, the highest of any quasi-stable nucleus) is the extremum of Wallace's own design axis and independently the corpus's rung-2 substrate. This is the best unplayed card in the electrogravitics deck — and it is worth being precise about *why* it survives where every other rotating-mass saucer claim (Carr, Searl, the Repulsine tonnage) does not: the 1990 gyroscope nulls (§3) bound the weight of *bulk* spinning mass; Wallace's variable is a *nuclear-spin polarization* nobody has ever put on a balance. The rerun tests an untested predicate, not a re-tested one. OPEN
2. **The hairpin bench.** The two classical force laws for current elements (Grassmann/Lorentz vs Ampère) agree exactly on every closed circuit and differ by a finite, cutoff-free **29%** on an open hairpin bridge closing through rails (5.35 vs 6.91 gram-force at 270 A — `crypto_oracle/universe_sim/ampere_tension_sim.py`). A supercap bank, a load cell, and a weekend settle a 200-year-old question about longitudinal forces — and the open-circuit/closed-circuit distinction is the same one that governs whether a craft can push on the lattice at all ("the craft is the bridge, the substrate is the rails"). DERIVED, bench-ready
3. **The sign-flip protocol as a community standard.** Most of this field's hundred-year noise floor exists because positives were reported without the §6 discriminator set. A drone company that publishes its balance design, its five controls, and its nulls — whatever the outcome — becomes the credible instrument this field has never had. That reputation is worth more than a marginal positive.
4. **The document hunt.** Three archival retrievals would move the §3 ledger more than any argument. One is now done: the SNCASO "Project Montgolfier" reports

(1957 narrative + 1959 final, Cornillon's own copy, archive.org) were read in the original on 2026-08-18 — see the §3 row and `MONTGOLFIER_REPORT_RETRIEVAL_2026-08-18.md`; the residue is a single 1 g·cm torsion datum on a $BaTiO_3$ article and nulls on everything else. Still open: the 1968 Northrop electro-aerodynamics paper (drag reduction, not thrust — but a real claimed AIAA item reported vanished). The 1952 Navy appraisal, by contrast, is *not* missing: it is ONR Pasadena File 24-185 (W. M. Cady, 15 Sep 1952, declassified 1 Oct 1952), circulating as a PDF, and it is the only calibrated measurement of a Brown saucer on record — see the ledger row above. Relatedly: Brown's *electrometer* records — the only tabulated year, 1937, was analysed by Cady, who found no solar and no lunar period and wrote that no test had been applied for the claimed *sidereal* period; the raw hourly table has not surfaced since — the §6 sidereal-folding discriminator adjudicates that claim mechanically if it does, and either outcome is informative (Townsend Brown §4, §9).

9. The Electrogravitics Claim

> 1. The force Townsend Brown measured in air is ion wind — conventional, useful, atmospheric, and not gravitational; and the *static* vacuum force is zero by measurement and by this framework's own evenness lemma. **MEASURED**
> 2. Every positive in the vacuum record is *transient* — flash, breakdown, pulse, high-K — which is the one regime the framework's force law allows and the one the published null program has not entered; its conventional candidate (arc-ejecta recoil, $\sim 10^{-5}$ N·s/C) has never been discriminated by impulse-per-coulomb, mass, chopper-vs-spark, or K-scaling (§1.1). **OPEN**
> 3. Every published, independently instrumented vacuum test of the Brown effect is a null, and the tightest bounds the static-capacitor coupling at $\sim 2\times10^{-10}$. The one claimed vacuum positive — Project Montgolfier, 1955–59 — has been read in the original: the Brown disc–wire article was null on the 1959 torsion balance and the single positive ($\approx$1 g·cm on a $BaTiO_3$-loaded capacitor at ±55 kV) carries no current or breakdown log; it stays open until a replication with a current-and-force log or the §6 bench speaks. **MEASURED** (published record) / **OPEN** (the French residual)
> 4. The mechanism exists unconditionally: electromagnetic fields curve spacetime (Einstein–Maxwell, $8\pi G/c^4$; exact lab-hardware solutions, Füzfa 2016). The companion framework derives a specific force law — F = C·κ·U_drive/L_grad — in which Brown's

geometry is the right shape and coherence is the proposed missing ingredient; weak-field gravity itself is closed by the underlying law with receipts. THEOREM (existence) / DERIVED (form)

5. That law's coupling constant C is the single open number. It has never been measured above zero. Hover requires $\sim 10^{-6}$ of full coherence; the bare incoherent floor is $\sim 10^{-43}$. OPEN
6. Hover, if the coupling exists, is a near-zero-power static lock; propellantless is not reactionless; symmetric drives give zero; over-unity is forbidden. DERIVED
7. The decisive experiment is drone-shop-sized: one asymmetric device, one vacuum balance, five controls (§6). A positive that survives them changes everything; a clean null is the best bound ever published. OPEN
8. No engineering claim in this lineage has survived audit to date — and this paper's own program is the audit it must survive. MEASURED

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- Documentary audit behind the §3 Montgolfier row and the §8 document hunt: `papers/MICHELS_BROWN_DOCUMENTARY_AUDIT_2026-08-13.md` (§1/§1a).

Changelog

- **2026-08-13** — draft 0.1, written and revised in one day: initial field guide; mechanism register raised to theorem level (Einstein–Maxwell / Füzfa 2016 exact solutions; three-layer claim structure); Biefeld–Brown retrodiction sim run against its July pre-registration and scored (all four frozen predictions, all three gates); Montgolfier documentary audit folded into the §3 ledger and the §8 document hunt; Buhler claim-decline recorded; coverage analysis of the published null program (the one untested cell) added to §4; over-unity boundary stated precisely.
- **2026-08-14** — draft 0.2: editorial pass — the day's addenda folded into the body prose; §2 restructured to lead with the Einstein–Maxwell theorem; audit-file provenance moved to References and this changelog. No claim, number, or tag changed.

- **2026-08-18** — draft 0.3: Otis Carr audit folded in (companion Otis T. Carr and the OTC-X1): §3 ledger gains the spinning-mass weight-change null row (Hayasaka–Takeuchi 1989 → Faller / Nitschke & Wilmarth / Quinn & Picard 1990) — a bound the ledger had been missing, and the one row that closes every "spin it and it gets light" saucer design at once — plus a Carr/Utron row; §7 dead-ends gains the Carr entry; §8 Wallace item now states why it survives that null (nuclear-spin polarization is an untested predicate, bulk spin is not); references extended. Townsend Brown companion pointers (added the same day) unchanged.

Earlier drafts of companion papers called the lattice coupling x_v; the current canonical object is the vertex-sharing strength $g \le 1/\sqrt{3}$ with substrate aggregate $C_{sub} \cdot |xv|^2$ kept as the engineering coefficient. This paper uses "the coupling, C" throughout and lets the companions own the formalism.